Dictated using Wispr Flow

For the lyrical only, I would recommend using the TF-IDF-weighted cosine similarity. When looking at the different positions, I was able to tell that the precision for the dot product was only about 9%. When we did the cosine, it was about 19.1%, and when we did the TF-IDF, it was about 21%. That's the reasoning behind choosing that, and it is also really just the strongest of the three. It fixes the biggest problem with the dot product, which only rewards songs that are simply longer or more repetitive.

When we compare to look at the top 10 songs, that's a very clear pattern that we can see. Some of the largest songs that the dot product had were close to 2,000 words, really. The songs that we saw in the cosine were words that were about 300, so it is a pretty clear indication that the dot product itself is looking for songs that are simply longer or more repetitive.

With regard to the TF-IDF, It fixes the issue of weighting down generic filler words, so words like "I" and "you" get a lower weighting. You get a higher rating for distinct or unique words, and then combining that with corrections is why the TF-IDF cosine performed as well.

I think a good example for this would be looking at the rankings for the song "What Are We Here For" The dot product matches the Interlude 5 and the "We Gonna Rock" song. These weren't songs that are similar, but the Interlude 5 is one of the longest song in the entire database, so it accumulates a large dot product purely because of how big it is. The "We Gonna Rock" song has about 10 unique words that are repeated constantly, so that also inflates the score just because there's a lot of repetition. Once we switch to similarity through cosine, the songs were not even there in the top, but they were actually replaced by songs that are shorter but also had specific, unique phrases. The lyrics are sort of a weak way to represent this anyway.

Even at the best baseline, we're only about to get 21% precision, meaning that 1 in 4 recommendations actually share the right genre. It's still not accurate, and that's the limitation that we see here. Two songs can be really different in terms of vocabulary, but they can also belong to the same genre. Also, two songs can share very common words, and they might not even be similar at all. This might not be an effective recommendation system for combining songs.

I also, in part two, tried to build my own system that was better by combining the TF-IDF cosine similarity with the counts correlation, by averaging each song's rank rather than averaging their scores. The idea behind it was that cosine and correlation capture two different signals. Using cosine on a TF-IDF its shows the distinct vocabulary, while correlation captures whether a song should rise or fall relative to its own baseline words used. Combining them should remove any individually misleading words, but this only turned out to be 0.5% better than the TF-IDF. This would probably not be the best approach either. Although it did do better, I don't think this is the best way to approach this. Since it still provides a bad recommendation, there's probably an alternative method that Spotify uses, which is why it's so good and why people use it. I think all these are not the best, but it gives us an understanding of what is happening in terms of how to find similar songs and how a company like Spotify may be doing it to find songs in similar genres. A proper way to do this is probably using lyrics information along with other parts of a song. Looking into the structure of the song like musical tempos along with human behavior would be really helpful. Now in this AI world I would also say maybe trying to understand the words and maybe even finding synonyms would be an added plus.